AGENTS' ROOM: NARRATIVE GENERATION THROUGH MULTI-STEP COLLABORATION

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ABSTRACT

Writing compelling fiction is a multifaceted process combining elements such as crafting a plot, developing interesting characters, and using evocative language. While large language models (LLMs) show promise for story writing, they currently rely heavily on intricate prompting, which limits their use. We propose AGENTS' ROOM, a generation framework inspired by narrative theory, that decomposes narrative writing into subtasks tackled by specialized agents. To illustrate our method, we introduce TELL ME A STORY[1], a high-quality dataset of complex writing prompts and human-written stories, and a novel evaluation framework designed specifically for assessing long narratives. We show that AGENTS' ROOM generates stories that are preferred by expert evaluators over those produced by baseline systems by leveraging collaboration and specialization to decompose the complex story writing task into tractable components. We provide extensive analysis with automated and human-based metrics of the generated output.

1 INTRODUCTION

Creating long-form content requires meticulous research, advanced planning, an engaging writing style, and the ability to craft stories that captivate. J.K. Rowling is claimed to have had most of the Harry Potter story planned out before she started writing. She knew there would be seven books, which characters would be important and how they would develop, and which key plot twists would serve the overall story. In addition, she carried out substantial research to create the fictional universe which provides the backdrop of the story. Breaking down a story into distinct sections is typical for longer narratives, with most stories boiling down to a few shared elements like exposition, rising action, climax, falling action, and resolution (Freytag, 1896; Pavis, 1998). Practical guides to writing successful screenplays (Cutting, 2016; Hauge, 2017) outline a similar structure, following the setup, the new situation, progress, complications and higher stakes, the final push, and the aftermath.

Large language models (LLMs) have demonstrated impressive writing capabilities (Yang et al., 2022; Nur Fitria, 2023; Shao et al., 2024; Bai et al., 2024), however, generating long-form content is still a challenge. Well-known problems include maintaining a consistent narrative, tone, or factual accuracy over extended stretches of text (Chakrabarty et al., 2024b; Wang et al., 2023b; Alabdulkarim et al., 2021; Balepur et al., 2023; Yamshchikov & Tikhonov, 2023), and showcasing a unique voice or humor that makes writing truly memorable. Despite displaying flashes of creativity, they often replicate patterns found in their training data, which hinders the generation of original concepts, plotlines, or phrasing. Added problems include the lack of datasets or benchmarks for long-form writing (Bai et al., 2024) and standardized evaluation criteria for assessing creative writing either by humans or machines (Chhun et al., 2022; 2024; Chakrabarty et al., 2024a).

Existing methods often rely on detailed prompts to guide the generation process (Yang et al., 2022; Xie et al., 2023), prompt chaining (Mirowski et al., 2023; Yang et al., 2022), and planning strategies (Yang et al., 2023; Lee et al., 2024) as a means of breaking down the complex writing task into more manageable components. In this paper, we conceptualize long-form writing as a multi-agent

[1]We release the dataset and metrics at: https://github.com/google-deepmind/tell_me_a_story

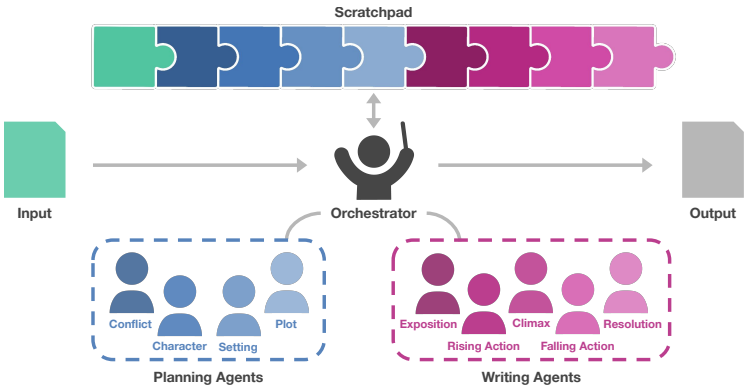

Figure 1: AGENTS' ROOM, a multi-agent framework for collaborative writing. A central orchestrator calls the individual agents and consolidates their contributions into the scratchpad. We color-code each piece of the scratchpad with the contributing agent's color.

collaboration problem. Rather than attempting a decomposition of the writing task within a *single* agent (Chen et al., 2023; Yao et al., 2024), we leverage collaboration among *multiple* agents, with specialized abilities (Talebirad & Nadiri, 2023; Zhang et al., 2024b; Han et al., 2024). We propose AGENTS' ROOM[2] (Figure 1), a generation paradigm which consists of two types of agents, namely *planning* and *writing* agents. Planning agents flesh out key components of the content but do not write the story as such. For example, a planning agent might specialize in character descriptions, whereas another might focus on the plot or central conflict. Writing agents are responsible for generating the final output text and are also specialized, e.g., one may focus on the introduction, and another on the conclusions. The two types of agents work collaboratively to complete the writing task, sharing and managing information through a *scratchpad* which maintains outputs from planning agents and makes them available to writing agents. An *orchestrator* is responsible for calling the agents in order depending on the task at hand.

Compared to single LLM-powered agents, this multi-agent approach offers several advantages:

- LLMs can be specialized into various distinct agents (e.g., zero-shot prompted or fine-tuned) performing a single function with high precision;
- it avoids well-known problems with lengthy and under-specified instructions which require multiple iterations to build context and fully define an appropriate solution;
- it can be applied to problems whose solution is not known beforehand, and results from exploring a vast research space or involves very long output (e.g., writing a book);
- it naturally lends itself to human-in-the loop automation where machine-based agents can be replaced with human ones when needed.

We formalize AGENTS' ROOM as a general writing framework and apply it to creative writing. Specifically, we focus on writing long-form stories (1,000-2,000 tokens), and create specialized agents drawing inspiration from narrative theory (e.g., Card 1999; Noble 1999; Pavis 1998). To evaluate our method, we introduce TELL ME A STORY, a new dataset of human-created writing prompts and fiction stories, and a novel evaluation framework designed for assessing multiple dimensions of story quality. Experimental results show that AGENTS' ROOM generates stories that are preferred (by humans and automatic metrics) over those produced by baseline systems which do not leverage collaboration or specialization.

2 RELATED WORK

Story Generation The advent of large pre-trained language models has provided a common framework for generating stories which sound fluent but often struggle with maintaining coherence and

[2] AGENTS' ROOM is very loosely modeled after writers' room, a collaborative space where writers, (usually of a television series), come together to write and refine scripts.

plausibility. Attempts to enhance coherence and control the trajectory of events often decompose the generation task into planning an outline or sketch, and then elaborating on it, e.g., by filling in descriptions and specific details of each story. Examples of intermediate plans include sequences of entities and their actions (Yao et al., 2019), outlines (Fan et al., 2019; Zhou et al., 2023; Wang et al., 2023a), plot structures (Goldfarb-Tarrant et al., 2020), and more elaborate descriptions including details about the setting of the story, its characters, and main plot points (Yang et al., 2022; 2023). Other work uses common sense knowledge to impose constraints on the characters and their interactions (Peng et al., 2022), ensemble-based models to render event sequences more plausible (Ammanabrolu et al., 2020), stylistic constraints (Kong et al., 2021), and twists through constrained decoding (Huang et al., 2023). These efforts have demonstrated that generating stories as a one-step process is challenging, and ultimately various interventions are required to improve overall story quality. Our work follows on from this realization, and breaks down the writing task into subtasks, undertaken by different agents who collaboratively plan and write a story. Collaborative writing is often used in academic or professional contexts to leverage the strengths and perspectives of various contributors, and has also been shown to enhance creativity (Barrett et al., 2021).

Using LLMs as tools to assist humans with writing stories is an active research area (Chakrabarty et al., 2024b; Mirowski et al., 2023; Ippolito et al., 2022). In our experiments, stories are written exclusively by models without humans in the loop. However, our framework is fairly general allowing for human-machine collaboration at various stages of content creation.

Multi-agent Systems LLM-based agents have recently shown robust reasoning and planning capabilities across various application domains (Zhao et al., 2023; Bubeck et al., 2023). Multi-agent systems involve multiple independent LLMs working together to solve complex tasks that are beyond the capability of any individual agent (Talebirad & Nadiri, 2023; Park et al., 2023; Han et al., 2024; Guo et al., 2024). The agents are typically specialized in different aspects of a problem or have different roles, allowing the system to approach tasks in a more coordinated, distributed, and modular way. LLM-based multi-agent systems have recently demonstrated promising results in multiple areas including software development (Hong et al., 2024), robotic tasks such as motion planning (Mandi et al., 2024), simulations of human behavior (Park et al., 2023; Hua et al., 2024), the creation of gaming enviroments (Hu et al., 2024), recommender systems (Zhang et al., 2024a), simulations of financial trading (Li et al., 2023), and policy making (Xiao et al., 2023). We are not aware of existing multi-agent frameworks for long-form writing. We draw inspiration from related work demonstrating that collaborative problem-solving improves LLM task-solving capabilities (Hao et al., 2023; Wang et al., 2024; Zhang et al., 2024b). Our agents each adopt a specialized writing subtask and communicate through a shared scratchpad (or memory) which allows to effectively recall and utilize contextually-relevant past knowledge. In our experiments, we predefine the number and type of agents best suited to our story writing task, rather than dynamically generate agents based on story content (Chen et al., 2024).

Evaluation Story evaluation is admittedly a challenging task for humans and machines. Human evaluation is usually considered as the gold standard, but it is expensive, time-consuming (Guan & Huang, 2020), and can be subjective. It also cannot capture diversity since a model that copies directly from the training set would potentially pass the human quality bar without displaying any generalization or creativity (Hashimoto et al., 2019). Automated evaluation metrics based on lexical overlap or semantic similarity between generated stories and their human references have been shown to correlate poorly with human judgements (Chhun et al., 2022). In this paper, we introduce an LLM-based evaluator (Liusie et al., 2023; Liu et al., 2024; Zheng et al., 2024; Bohnet et al., 2024) to perform side-by-side comparisons of system outputs which correlates with human judgements. Inspired by recent proposals on how to assess human creativity (Chakrabarty et al., 2024a), we distill the story evaluation task into a few dimensions (e.g., plot, language use) which humans and machines can judge reliably.

3 AGENTS' ROOM

In this section, we formalize AGENTS' ROOM, the proposed multi-agent framework for collaborative writing. Given a complex writing task x, we generate output y, by decomposing the writing process into multiple subtasks tackled by specialized agents. The full AGENTS' ROOM framework is summarized in Algorithm 1 and explained below.